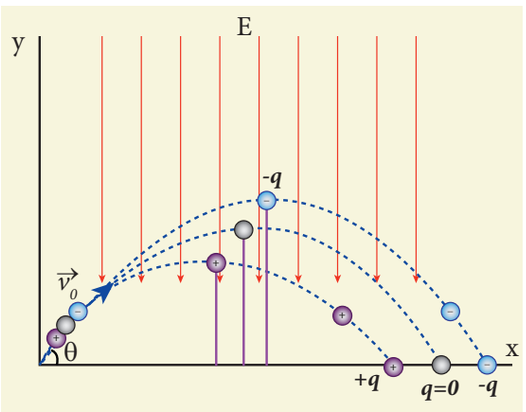
range, maximum height and time of flight. In fact we can simply replace g by $g+\frac{qE}{m}$ in the usual expressions of range, maximum height and time of flight of a projectile.

	Without charge	With the charge +q
Time of flight T	$\frac{2v_\circ \sin\theta}{g}$	$\frac{2v_\circ \sin\theta}{\left(g+\frac{qE}{m}\right)}$
Maximum height h_{max}	$\frac{v_\circ^2 \sin^2\theta}{2g}$	$\frac{v_\circ^2 \sin^2\theta}{2\left(g+\frac{qE}{m}\right)}$
Range R	$\frac{v_\circ^2 \sin 2\theta}{g}$	$\frac{v_\circ^2 \sin 2\theta}{\left(g+\frac{qE}{m}\right)}$

Note that the time of flight, maximum height, range are all inversely proportional to the acceleration of the object. Since $\left(g+\frac{qE}{m}\right) > g$ for charge $+q$, the quantities T, h_{max}, and R will decrease when compared to the motion of an object of mass m and zero net charge. Suppose the charge is $-q$, then $\left(g-\frac{qE}{m}\right) < g$, and the quantities T, h_{max} and R will increase. Interestingly the trajectory is still parabolic as shown in the figure.



1.7.4 Dielectrics or insulators

A dielectric is a non-conducting material and has no free electrons. The electrons in a dielectric are bound within the atoms. Ebonite, glass and mica are some examples of dielectrics. When an external electric field is applied, the electrons are not free to move anywhere but they are realigned in a specific way. A dielectric is made up of either polar molecules or non-polar molecules.

Non-polar molecules

A non-polar molecule is one in which centres of positive and negative charges coincide. As a result, it has no permanent dipole moment. Examples of non-polar molecules are hydrogen (H_2), oxygen (O_2), and carbon dioxide (CO_2) etc.

When an external electric field is applied, the centres of positive and negative charges are separated by a small distance which induces dipole moment in the direction of the external electric field. Then the dielectric is said to be polarized by an external electric field. This is shown in Figure 1.48.

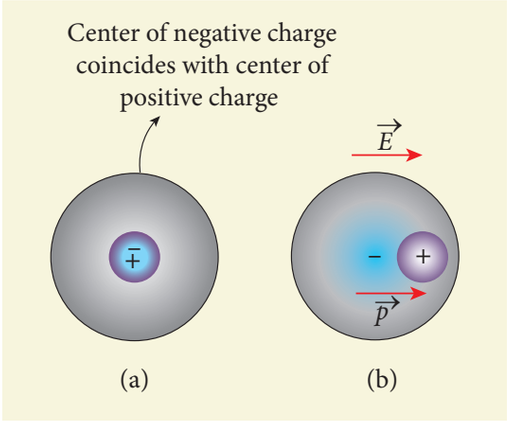


Figure 1.48 Non polar molecules (a) without external field (b) with the external field

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